

IMAT QUESTION BANK: Chemistry (Medium-Hard)

65 questions · 108 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · Question bank

Section A - Organic chemistry and isomerism (Q1-13)

1. D-glucose and D-galactose are both aldohexoses with the same bonding order; their Fischer projections differ only at carbon 4, where the OH points right in glucose and left in galactose. Which term most precisely classifies the pair?
- A) They are enantiomers, because the OH at C-4 is on opposite sides in the two sugars.
 - B) They are identical compounds drawn as different Fischer projections.
 - C) They are epimers, because they differ in configuration at exactly one chiral center, C-4.
 - D) They are constitutional isomers, because in galactose the C-4 OH is bonded to a different carbon.
 - E) They are anomers, because any two aldohexoses that differ at a single carbon are anomers.
2. A synthesis produces a molecule with the same bonding order as (2R,3S,4R)-hexane-2,3,4-triol but with configurations (2S,3R,4S) at its three chiral centers. Which term most precisely classifies the relationship between the two molecules?
- A) They are constitutional isomers, because the OH groups occupy different positions.
 - B) They are epimers, because several chiral centers change at the same time.
 - C) They are diastereomers, because some of the stereocenters are inverted.
 - D) They are enantiomers, because every chiral center is inverted, giving two non-superimposable mirror images.
 - E) They are identical molecules, because they share the same molecular formula.
3. In water, D-glucose cyclises and exists as two forms that differ only at carbon 1: in the alpha form the new OH points down from the ring, in the beta form it points up, while the configurations at carbons 2 to 5 are identical. Which term most precisely classifies the two forms?
- A) Conformational isomers, interconvertible by rotation about a single bond.
 - B) Anomers, because they differ in configuration at the anomeric carbon, C-1.
 - C) Constitutional isomers, because the OH is attached to the ring in two different ways.
 - D) Enantiomers, because the anomeric carbon is the most important stereocenter.
 - E) Identical compounds, because the two forms have the same molecular formula.
4. Butane-2,3-diol exists in stereoisomeric forms: one has the configuration (2R,3R) and another has (2R,3S). Which term most precisely classifies these two forms?
- A) They are conformational isomers, because they interconvert by bond rotation at room temperature.
 - B) They are constitutional isomers, because an OH group has moved from C-2 to C-3.
 - C) They are enantiomers, because one form contains an R descriptor and the other an S descriptor.
 - D) They are identical, because rotation about the C-2 to C-3 bond superimposes them.
 - E) They are diastereomers, because they are stereoisomers that are not mirror images of each other.
5. Compound X is $\text{CH}_3\text{OCH}_2\text{CH}_3$ (methoxyethane) and compound Y is $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (propan-1-ol); both have molecular formula $\text{C}_3\text{H}_8\text{O}$. Which statement is correct?
- A) They are constitutional isomers with different functional groups: an ether and an alcohol.
 - B) They are enantiomers, because the alcohol is a chiral molecule.
 - C) They are diastereomers, because their physical properties differ.
 - D) They are the same compound drawn differently, since both have formula $\text{C}_3\text{H}_8\text{O}$.
 - E) They are both ethers, because both contain a C-O single bond.

6. D-glucose is an aldohexose with an aldehyde group at C-1, while D-fructose is a ketohexose with a ketone group at C-2. Both have molecular formula $C_6H_{12}O_6$. Which term most precisely classifies the pair?
- A) Epimers, because they differ at one carbon atom.
 - B) Stereoisomers, because only the position of the carbonyl differs.
 - C) Constitutional isomers, because the same atoms are joined in a different order.
 - D) Anomers, because the carbonyl sits at C-2 in one sugar and C-1 in the other.
 - E) Enantiomers, because both are hexoses with the same formula.
7. The structure $CH_3CH_2CONH_2$ is named:
- A) Propanamide
 - B) Propanenitrile
 - C) Propanoic acid
 - D) Propanamine
 - E) Methyl ethanoate
8. Oil of wintergreen (methyl 2-hydroxybenzoate) consists of a benzene ring bearing an OH group attached directly to the ring, plus a $COOCH_3$ group joined through the carbonyl carbon. Which two functional groups does the molecule contain?
- A) A phenol and an amide
 - B) An ether and a ketone
 - C) A carboxylic acid and an ester
 - D) A phenol and an ester
 - E) An alcohol and a carboxylic acid
9. The structure $CH_3CH_2OCH_2CH_3$ is named:
- A) Butanal B) Ethoxyethane C) Butan-1-ol D) Butan-2-one E) Butanoic acid
10. Which of the following compounds is a ketone and contains exactly four carbon atoms?
- A) Butanal, $CH_3CH_2CH_2CHO$
 - B) Propanone, CH_3COCH_3
 - C) Butan-2-ol, $CH_3CH(OH)CH_2CH_3$
 - D) Butanoic acid, $CH_3CH_2CH_2COOH$
 - E) Butan-2-one, $CH_3COCH_2CH_3$
11. An aldohexose is drawn as a Fischer projection with the aldehyde at the top as C-1. The OH groups at C-2, C-3 and C-4 point right, right and left respectively, and the OH at C-5, the chiral center farthest from the aldehyde, points left. How is the sugar classified?
- A) It is an L-sugar, because the OH on the stereocenter farthest from the aldehyde points left.
 - B) It is meso, because left- and right-pointing OH groups balance two against two.
 - C) It is a D-sugar, because the aldehyde group is drawn at the top of the projection.
 - D) It is a D-sugar, because D/L is fixed by the OH closest to the aldehyde group.
 - E) It is a D-sugar, because two of its four OH groups point right.
12. Maltose and cellobiose are both disaccharides of two glucose units with molecular formula $C_{12}H_{22}O_{11}$. Maltose joins its units with an α -1,4 glycosidic bond and cellobiose with a β -1,4 bond. Human enzymes hydrolyse maltose efficiently but cannot digest cellobiose. Which statement explains this best?
- A) Maltose is digested because it is a reducing sugar, whereas cellobiose is non-reducing.
 - B) Cellobiose passes through the gut unchanged because no enzyme can release glucose from a beta bond.
 - C) Cellobiose cannot be digested because its molecular formula differs from that of maltose.
 - D) Human enzymes fit the α -1,4 geometry of maltose but not the β -1,4 geometry of cellobiose, so cellobiose is not hydrolysed.
 - E) Cellobiose is indigestible because beta glycosidic bonds are chemically stronger than alpha bonds and cannot be broken in water.

13. In sucrose, the glycosidic bond links the anomeric carbon C-1 of glucose to the anomeric carbon C-2 of fructose. In maltose, the glycosidic bond uses the anomeric carbon of only one of its two glucose units. Which statement is correct?

- A)** Sucrose is reducing because fructose is a ketose, and ketoses are always stronger reducing agents.
- B)** Sucrose is non-reducing because both anomeric carbons are locked inside its glycosidic bond, while maltose is reducing because one anomeric carbon stays free.
- C)** Maltose is non-reducing, because it is built only from glucose units.
- D)** Sucrose is reducing, because the fructose ring can still open at C-2 while glycosidically bonded.
- E)** Both sugars are non-reducing, because any glycosidic bond blocks reduction.

Section B - Stoichiometry, moles and gas laws (Q14-27)

14. How many molecules are present in 4.4 g of propane, C_3H_8 ? (H = 1, C = 12; Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

- A)** 1.204×10^{23} **B)** 3.01×10^{22} **C)** 6.02×10^{21} **D)** 6.02×10^{22} **E)** 6.02×10^{23}

15. How many hydrogen atoms are present in 0.25 mol of water, H_2O ? (Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

- A)** 1.204×10^{24} **B)** 1.505×10^{23} **C)** 3.01×10^{23} **D)** 6.02×10^{23} **E)** 9.03×10^{23}

16. What mass of ammonia, NH_3 , contains the same number of molecules as 4.4 g of carbon dioxide, CO_2 ? (C = 12, O = 16, N = 14, H = 1)

- A)** 1.7 g **B)** 3.4 g **C)** 4.4 g **D)** 8.5 g **E)** 17 g

17. What volume does 8.8 g of carbon dioxide, CO_2 , occupy at STP? (C = 12, O = 16; molar volume at STP = $22.4 \text{ dm}^3/\text{mol}$)

- A)** 4.48 dm^3 **B)** 8.8 dm^3 **C)** 11.2 dm^3 **D)** 22.4 dm^3 **E)** 44.8 dm^3

18. How many molecules are present in 560 cm^3 of an ideal gas at STP? ($1 \text{ dm}^3 = 1000 \text{ cm}^3$; molar volume at STP = $22.4 \text{ dm}^3/\text{mol}$; Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

- A)** 1.505×10^{22} **B)** 1.505×10^{24} **C)** 1.505×10^{25} **D)** 6.02×10^{21} **E)** 6.02×10^{23}

19. What is the mass of 4.48 dm^3 of oxygen gas, O_2 , measured at STP? (O = 16; molar volume at STP = $22.4 \text{ dm}^3/\text{mol}$)

- A)** 1.6 g **B)** 3.2 g **C)** 6.4 g **D)** 16 g **E)** 32 g

20. 10.0 g of calcium carbonate ($M = 100 \text{ g/mol}$) is treated with 100 cm^3 of 1.0 mol/dm^3 hydrochloric acid: $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$. What volume of CO_2 is released, measured at STP? ($1 \text{ dm}^3 = 1000 \text{ cm}^3$; molar volume at STP = $22.4 \text{ dm}^3/\text{mol}$)

- A)** 0.56 dm^3 **B)** 1.12 dm^3 **C)** 2.24 dm^3 **D)** 11.2 dm^3 **E)** 22.4 dm^3

21. In theory, 9.2 g of ethanol ($M = 46 \text{ g/mol}$) dehydrates completely to ethene ($M = 28 \text{ g/mol}$): $C_2H_5OH \rightarrow C_2H_4 + H_2O$. The actual mass of ethene collected is 4.2 g. What is the percentage yield?

- A)** 25% **B)** 56% **C)** 75% **D)** 100% **E)** 133%

22. A diver exhales an air bubble of volume 2.0 dm^3 at a depth where the pressure is 400 kPa. The temperature stays constant as the bubble rises to the surface, where the pressure is 100 kPa. What is the volume of the bubble at the surface?

- A)** 0.5 dm^3 **B)** 1.0 dm^3 **C)** 2.0 dm^3 **D)** 4.0 dm^3 **E)** 8.0 dm^3

23. A rigid sealed canister holds a fixed amount of gas at 27°C and 100 kPa. It is heated until the gas reaches 327°C . What is the new pressure inside the canister?

- A)** 50 kPa **B)** 100 kPa **C)** 109 kPa **D)** 200 kPa **E)** 300 kPa

24. A gas occupies 9.0 dm^3 at 27°C and 100 kPa. What volume does it occupy at 127°C and 200 kPa?

- A)** 4.5 dm^3 **B)** 6.0 dm^3 **C)** 12 dm^3 **D)** 13.5 dm^3 **E)** 24 dm^3

- 25.** A sealed vessel contains 0.20 mol of nitrogen gas and 0.30 mol of oxygen gas at constant temperature, and the total pressure is 150 kPa. What is the partial pressure of the oxygen?
A) 45 kPa **B)** 60 kPa **C)** 75 kPa **D)** 90 kPa **E)** 150 kPa
- 26.** 100 cm³ of 0.80 mol/dm³ hydrochloric acid is mixed with 400 cm³ of 0.20 mol/dm³ hydrochloric acid. Assuming the volumes are additive, what is the concentration of the resulting solution?
A) 0.16 mol/dm³ **B)** 0.20 mol/dm³ **C)** 0.32 mol/dm³ **D)** 0.50 mol/dm³ **E)** 0.80 mol/dm³
- 27.** A solution is prepared by diluting 25 cm³ of a 2.0 mol/dm³ solution to a concentration of 0.10 mol/dm³. What is the final volume of the diluted solution?
A) 25 cm³ **B)** 250 cm³ **C)** 450 cm³ **D)** 500 cm³ **E)** 1250 cm³

Section C - Solutions, acids and bases (Q28-40)

- 28.** What is the pH of a 2.0×10^{-4} mol/dm³ solution of hydrochloric acid, a strong acid that is fully ionised in water?
A) 2.7 **B)** 3.3 **C)** 3.7 **D)** 4.0 **E)** 10.3
- 29.** What is the pOH of a 5.0×10^{-2} mol/dm³ solution of nitric acid, a strong acid that is fully ionised in water? ($K_w = 1.0 \times 10^{-14}$ at 25 °C)
A) 1.3 **B)** 2.7 **C)** 12.3 **D)** 12.7 **E)** 13.7
- 30.** Strontium hydroxide, Sr(OH)₂, is a strong base that is fully dissociated in water. What is the pH of a 5.0×10^{-4} mol/dm³ solution of Sr(OH)₂? ($K_w = 1.0 \times 10^{-14}$ at 25 °C)
A) 3.0 **B)** 10.7 **C)** 11.0 **D)** 11.7 **E)** 12.0
- 31.** Four solutions are prepared at 25 °C ($K_w = 1.0 \times 10^{-14}$) from fully ionised solutes: P is 1.0×10^{-3} mol/dm³ HCl, Q is 5.0×10^{-3} mol/dm³ HCl, R is 1.0×10^{-3} mol/dm³ NaOH and S is 1.0×10^{-3} mol/dm³ Ba(OH)₂. Which option places the four solutions in order of increasing pH?
A) Q, P, S, R **B)** P, Q, R, S **C)** S, R, Q, P **D)** P, Q, S, R **E)** Q, P, R, S
- 32.** Pure water is heated from 25 °C to 60 °C. As the temperature rises, K_w increases, and the pH of pure water at 60 °C is measured to be 6.5. Which of the following statements is correct?
A) The water is basic, because heating pushes the hydroxide concentration above the hydrogen ion concentration.
B) The water is acidic, because a pH below 7 always means more H₃O⁺ than OH⁻.
C) The water is neutral, because K_w keeps the value 1.0×10^{-14} at every temperature.
D) The water is acidic, because heating destroys hydroxide ions faster than hydrogen ions.
E) The water is neutral, because [H⁺] and [OH⁻] stay equal while both become larger than at 25 °C.
- 33.** At 10 °C the ionic product of water has fallen to $K_w = 3.0 \times 10^{-15}$ mol² dm⁻⁶. What is the hydroxide ion concentration, [OH⁻], in pure water at 10 °C?
A) 1.0×10^{-7} mol/dm³
B) 1.7×10^{-8} mol/dm³
C) 3.0×10^{-15} mol/dm³
D) 5.5×10^{-8} mol/dm³
E) 9.0×10^{-8} mol/dm³
- 34.** 10 cm³ of 0.10 mol/dm³ hydrochloric acid, fully ionised, is diluted with water to a final volume of 1.0 dm³. By how much does the pH of the solution change?
A) It stays the same, because the amount of acid is unchanged.
B) It reaches pH 7, because a hundredfold dilution turns any strong acid neutral.
C) It falls by 2 units.
D) It rises by 1 unit.
E) It rises by 2 units.

- 35.** What volume of water must be added to 20 cm³ of 1.0×10^{-2} mol/dm³ hydrochloric acid, fully ionised, to obtain a solution of pH 4.0 at 25 °C?
A) 180 cm³ **B)** 198 cm³ **C)** 200 cm³ **D)** 1980 cm³ **E)** 2000 cm³
- 36.** Which of the following mixtures, made up with pure water as solvent, can act as a buffer solution?
A) 100 cm³ of 0.10 mol/dm³ ethanoic acid mixed with 100 cm³ of 0.10 mol/dm³ sodium hydroxide.
B) 100 cm³ of 0.10 mol/dm³ ethanoic acid mixed with 50 cm³ of 0.10 mol/dm³ sodium hydroxide.
C) 100 cm³ of 0.10 mol/dm³ hydrochloric acid mixed with 50 cm³ of 0.10 mol/dm³ sodium hydroxide.
D) A 0.10 mol/dm³ solution of ethanoic acid alone in water.
E) 100 cm³ of 0.10 mol/dm³ hydrochloric acid mixed with 100 cm³ of 0.10 mol/dm³ sodium chloride.
- 37.** A buffer solution contains comparable amounts of ethanoic acid and sodium ethanoate and has pH 4.8. An equal volume of pure water is added to it. Which of the following statements is correct?
A) The pH rises by about one unit because both concentrations fall by a factor of two.
B) The pH falls because the acid becomes effectively stronger on dilution.
C) The buffer is destroyed, because dilution pushes the dissociation of the acid completely towards the ions.
D) The pH and the buffer capacity are both unchanged, because a buffer resists any change made to it.
E) The pH stays close to 4.8 because the ratio of ethanoate to ethanoic acid is unchanged, but the buffer capacity falls.
- 38.** Which of the following statements about the behaviour of salts dissolved in water is correct?
A) Ammonium chloride gives a basic solution, because the ammonium ion accepts a proton from water.
B) Sodium ethanoate gives a basic solution, because the ethanoate ion accepts a proton from water and releases OH⁻.
C) Sodium ethanoate gives an acidic solution, because the sodium ion donates protons to water.
D) Sodium chloride gives a basic solution, because the chloride ion accepts protons from water.
E) Ammonium nitrate gives a basic solution, because the nitrate ion accepts protons from water.
- 39.** A drip bag holds 500 cm³ of glucose solution containing 9.0 g of glucose ($M = 180$ g/mol); the label reads 1.8 % w/v, meaning 1.8 g of glucose per 100 cm³ of solution. What is the molarity of the glucose?
A) 0.010 mol/dm³ **B)** 0.050 mol/dm³ **C)** 0.100 mol/dm³ **D)** 0.180 mol/dm³ **E)** 1.80 mol/dm³
- 40.** At 25 °C the solubility of calcium hydroxide in water is 1.85 g per dm³ of saturated solution, and the dissolved salt is fully dissociated. ($M(\text{Ca}(\text{OH})_2) = 74$ g/mol; $K_w = 1.0 \times 10^{-14}$ at 25 °C; for dilute aqueous solutions 1 ppm is 1 mg per dm³.) Which of the following statements is correct?
A) The solubility is 0.025 mol/dm³ and the pH of the saturated solution is 12.70.
B) The solubility is 0.050 mol/dm³ and the pH of the saturated solution is 13.30.
C) The solubility is 0.050 mol/dm³ and the pH of the saturated solution is 12.70.
D) The solubility is 0.025 mol/dm³ and the calcium ion concentration is 1850 ppm.
E) The solubility is 0.025 mol/dm³ and the pH of the saturated solution is 11.30.

Section D - Redox and oxidation numbers (Q41-49)

- 41.** What is the oxidation number of the carbon atom in methanal, HCHO?
A) -2 **B)** 0 **C)** +1 **D)** +2 **E)** +4

- 42.** In the thiosulfate ion, $\text{S}_2\text{O}_3^{2-}$, the structure has a central sulfur atom bonded to three oxygen atoms and to one terminal sulfur atom, as if one oxygen of a sulfate unit had been replaced by sulfur. Which of the following statements about oxidation numbers in this ion is correct?
- A)** The average oxidation number of sulfur is +2, with the central sulfur at +5 and the terminal sulfur at -1.
 - B)** The average oxidation number of sulfur is +4, because each sulfur behaves as in the sulfate ion.
 - C)** Both sulfur atoms have oxidation number +2, because the charge of the ion is -2.
 - D)** The terminal sulfur has oxidation number -2 and the central sulfur +4, exactly as in sulfate.
 - E)** Oxidation numbers cannot be assigned, because the bond between identical atoms makes all the numbers undefined.
- 43.** Which of the following statements about oxidation numbers in oxoanions is correct?
- A)** Sulfur is +7 in SO_4^{2-} and nitrogen is +4 in NO_2^- .
 - B)** Sulfur is +6 in SO_4^{2-} and nitrogen is +5 in NO_2^- .
 - C)** Sulfur is +4 in SO_4^{2-} and nitrogen is +3 in NO_2^- .
 - D)** Sulfur is 0 in SO_4^{2-} because the oxygens carry the whole charge, and nitrogen is +3 in NO_2^- .
 - E)** Sulfur is +6 in SO_4^{2-} and nitrogen is +3 in NO_2^- .
- 44.** Ethanol, $\text{CH}_3\text{CH}_2\text{OH}$, is oxidised to ethanal, CH_3CHO . Which of the following statements is correct?
- A)** The carbon bearing the OH group goes from oxidation number -1 to +1, so ethanol is oxidised.
 - B)** Both carbon atoms rise from -3 to +1, so ethanol is oxidised.
 - C)** The oxidation number of oxygen rises from -2 to 0, so oxygen is oxidised.
 - D)** The average oxidation number of carbon is unchanged, so no redox change occurs.
 - E)** The carbon bearing the OH group goes from oxidation number +1 to -1, so ethanol is reduced.
- 45.** In aqueous solution, $2\text{FeCl}_3 + \text{SnCl}_2 \rightarrow 2\text{FeCl}_2 + \text{SnCl}_4$. Which species acts as the oxidising agent?
- A)** FeCl_2 , because iron ends at oxidation number +2.
 - B)** FeCl_3 , because Fe^{3+} is reduced to Fe^{2+} .
 - C)** Cl^- , because it keeps oxidation number -1 in every compound.
 - D)** SnCl_2 , because tin is reduced from +2 to +4.
 - E)** SnCl_4 , because tin is oxidised from +2 to +4.
- 46.** In the laboratory preparation of chlorine, MnO_2 reacts with concentrated hydrochloric acid: $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$. Which of the following statements is correct?
- A)** HCl is the oxidising agent, because it supplies the hydrogen ions.
 - B)** MnO_2 is the reducing agent, because manganese goes from +4 to +2.
 - C)** Chlorine gas is the reducing agent, because it is the oxidation product.
 - D)** No oxidation occurs, because no oxygen is transferred between the reactants.
 - E)** MnO_2 is the oxidising agent and the chloride ion is the reducing agent.
- 47.** Chlorine gas is bubbled into cold dilute sodium hydroxide: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$. Which of the following statements is correct?
- A)** Chlorine is oxidised only, and the hydroxide ion is the oxidising agent.
 - B)** Chlorine is both oxidised and reduced, while oxygen keeps oxidation number -2 throughout.
 - C)** The chloride ion is oxidised, because chlorine gas gains electrons.
 - D)** Oxygen is oxidised from -2 to 0 in the water that forms.
 - E)** Chlorine is reduced only, and the hydroxide ion is the reducing agent.
- 48.** Sulfite ions, SO_3^{2-} , are oxidised to sulfate ions, SO_4^{2-} , in acidic solution. When this change is written as a balanced half-equation, how many electrons appear on the right-hand side?
- A) 1 B) 2 C) 3 D) 4 E) 6**

49. In acidic solution, manganate(VII) ions, MnO_4^- , oxidise iron(II) ions, Fe^{2+} , to Fe^{3+} , the manganese ending as Mn^{2+} . In the balanced ionic equation, how many electrons are transferred per MnO_4^- ion, and what coefficient does Fe^{2+} carry?

- A)** 3 electrons and 3 Fe^{2+}
- B)** 4 electrons and 4 Fe^{2+}
- C)** 5 electrons and 1 Fe^{2+}
- D)** 5 electrons and 5 Fe^{2+}
- E)** 7 electrons and 7 Fe^{2+}

Section E - Bonding, polarity and intermolecular forces (Q50-59)

50. Butane (C_4H_{10} , Mr 58), propanone ($\text{C}_3\text{H}_6\text{O}$, Mr 58) and propan-1-ol ($\text{C}_3\text{H}_8\text{O}$, Mr 60) boil at approximately 0 °C, 56 °C and 97 °C, in some order. Which assignment is correct?

- A)** Butane 0 °C, propan-1-ol 56 °C, propanone 97 °C.
- B)** Butane 56 °C, propanone 97 °C, propan-1-ol 0 °C.
- C)** Butane 0 °C, propanone 56 °C, propan-1-ol 97 °C.
- D)** Propanone 0 °C, butane 56 °C, propan-1-ol 97 °C.
- E)** Propan-1-ol 0 °C, propanone 56 °C, butane 97 °C.

51. Butan-1-ol, butan-2-ol and 2-methylpropan-2-ol are isomers that all contain an O-H group. Their boiling points are approximately 118 °C, 100 °C and 82 °C respectively. Which statement correctly explains this trend?

- A)** Branching lowers the boiling point because it weakens the O-H bonds inside the molecules.
- B)** All three have the same boiling point, because hydrogen bonding dominates in each of them.
- C)** The boiling points differ because the three isomers contain different functional groups.
- D)** 2-methylpropan-2-ol has the highest boiling point, because its compact shape packs the molecules closest together.
- E)** Butan-1-ol has the highest boiling point, because its least branched chain gives the greatest molecular surface contact and so the strongest London forces, which act in addition to the hydrogen bonding present in all three isomers.

52. Hydrogen fluoride (Mr 20) boils at approximately 20 °C, while hydrogen chloride (Mr 36.5) boils at approximately -85 °C. Which statement explains this comparison?

- A)** HF molecules hydrogen bond to one another, a force much stronger than the dipole-dipole and London forces between HCl molecules, and this outweighs the higher molar mass of HCl.
- B)** HCl has the higher molar mass, so its London forces should make it boil higher, and the quoted temperatures cannot both be right.
- C)** The H-F covalent bond is stronger than the H-Cl covalent bond, and covalent bond strength sets the boiling point.
- D)** HF is ionic while HCl is covalent, and ionic compounds boil much higher.
- E)** HCl is non-polar, so only very weak London forces act between its molecules, while HF is also non-polar but lighter.

53. Which of the following statements is correct?

- A)** Water is non-polar, because its two identical O-H bonds pull equally in opposite directions.
- B)** Carbon dioxide is polar, because each C=O bond is strongly polar.
- C)** Boron trifluoride is non-polar, because its three identical B-F bond dipoles cancel in the trigonal planar arrangement.
- D)** Ammonia is non-polar, because the lone pair on nitrogen makes the molecule symmetrical.
- E)** Any molecule that contains polar bonds must be polar overall.

- 54.** Carbon tetrachloride, CCl_4 , chloroform, CHCl_3 , and dichloromethane, CH_2Cl_2 , are all tetrahedral molecules. Which statement about their polarity is correct?
- A)** All three are polar, because every tetrahedral molecule containing polar bonds is polar overall.
 - B)** The polarity of each molecule is decided only by its molar mass, so the heaviest molecule is the most polar.
 - C)** CCl_4 is the most polar of the three, because it contains the greatest number of C-Cl dipoles.
 - D)** All three are non-polar, because the tetrahedral shape of each molecule cancels every bond dipole.
 - E)** CCl_4 is non-polar because its four identical C-Cl dipoles cancel, while CHCl_3 and CH_2Cl_2 are polar because the C-H bonds do not balance the C-Cl dipoles.
- 55.** Carbon dioxide and sulfur dioxide both contain polar bonds to oxygen, yet CO_2 is non-polar and SO_2 is polar. Why?
- A)** CO_2 is linear, so its two C=O bond dipoles point in opposite directions and cancel, while SO_2 is bent, so its two S=O dipoles add to give a net dipole.
 - B)** Both molecules are linear, but only SO_2 has a lone pair on the central atom strong enough to create a dipole.
 - C)** SO_2 is bent, which makes its S=O bonds polar, while the linear shape of CO_2 makes its C=O bonds non-polar.
 - D)** SO_2 is polar because its molecules are heavier than CO_2 molecules and attract each other more strongly.
 - E)** The C=O bonds in CO_2 are non-polar, because carbon and oxygen have almost equal electronegativities.
- 56.** Four unknown substances were tested, with these results. P: melting point 801°C , conducts electricity only when molten or dissolved in water. Q: melting point -95°C , never conducts electricity. R: melting point 1085°C , conducts electricity as a solid. S: melting point 1710°C , never conducts electricity, even when molten. Which statement correctly identifies one of the substances?
- A)** R is metallic, because conduction in the solid state requires mobile delocalised electrons.
 - B)** Q is ionic, because ionic bonds are weak and break at a low temperature.
 - C)** S is ionic, because its very high melting point proves strong ionic bonds.
 - D)** P is metallic, because it conducts electricity when molten.
 - E)** P is a simple molecular substance, because molecular substances dissolve well in water.
- 57.** Sodium chloride melts at 801°C while iodine melts at 114°C , although iodine has the larger molar mass. Which statement explains the comparison?
- A)** Iodine melts easily because the covalent bonds between its molecules are weak.
 - B)** Sodium chloride has the higher melting point because its molar mass is greater than that of iodine.
 - C)** Melting sodium chloride means overcoming strong electrostatic attractions between oppositely charged ions in a giant lattice, while melting iodine only overcomes weak London forces between I_2 molecules, and the I-I covalent bonds stay intact.
 - D)** Sodium chloride contains covalent bonds, which are stronger than the covalent bonds inside iodine molecules.
 - E)** Both solids are giant lattices, and iodine simply has a lattice built from lighter particles.
- 58.** Use these average bond energies: H-H 436 kJ/mol , Cl-Cl 242 kJ/mol , H-Cl 431 kJ/mol . For the reaction $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$, the enthalpy change is approximately:
- A)** -368 kJ/mol **B)** -247 kJ/mol **C)** -184 kJ/mol **D)** $+184\text{ kJ/mol}$ **E)** $+247\text{ kJ/mol}$

- 59.** The thermal decomposition of calcium carbonate, $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$, has $\Delta H = +178 \text{ kJ/mol}$. Which statement is correct?
- A)** Bonds formed in the products are stronger than bonds broken in the reactants, which explains the positive ΔH .
 - B)** The products are lower in enthalpy than the reactants, so heat is released to the surroundings.
 - C)** At a sufficiently high temperature ΔH becomes negative, so the decomposition turns exothermic.
 - D)** The positive sign means energy is released by the system, so the reaction is exothermic.
 - E)** The products are higher in enthalpy than the reactants, and the difference is absorbed as heat from the surroundings.

Section F - Kinetics and catalysis (Q60-65)

- 60.** For reactions in dilute solution at a fixed temperature, the units of reaction rate are always $\text{mol}/(\text{dm}^3 \cdot \text{s})$. Which statement about rate constants is correct?
- A)** A zero-order rate constant has the units s^{-1} .
 - B)** A first-order rate constant has the units s^{-1} , and a second-order rate constant has the units $\text{dm}^3/(\text{mol} \cdot \text{s})$.
 - C)** A first-order rate constant has the units $\text{mol}/(\text{dm}^3 \cdot \text{s})$.
 - D)** Every rate constant has the same units as the rate itself, $\text{mol}/(\text{dm}^3 \cdot \text{s})$, whatever the order of reaction.
 - E)** A second-order rate constant has the units $\text{mol}/(\text{dm}^3 \cdot \text{s})$.
- 61.** A drug is removed from the bloodstream by first-order kinetics with a half-life of 4 hours. Starting from an initial dose D , the amount still present 12 hours after administration is:
- A)** 1/16 of D **B)** 1/8 of D **C)** 1/6 of D **D)** 1/4 of D **E)** 1/2 of D
- 62.** For the reaction $\text{A} + \text{B} \rightarrow \text{products}$, kept at constant temperature: doubling $[\text{A}]$ while holding $[\text{B}]$ constant doubles the rate, and doubling $[\text{B}]$ while holding $[\text{A}]$ constant increases the rate fourfold. Which statement is correct?
- A)** The reaction is second order overall, and k has the units $\text{dm}^3/(\text{mol} \cdot \text{s})$.
 - B)** The rate equation is $\text{rate} = k[\text{A}][\text{B}]^2$, the overall order is three, and k has the units $\text{dm}^3/(\text{mol} \cdot \text{s})$.
 - C)** The reaction is fourth order overall, because the fourfold increase in the second experiment means the exponent of B is three.
 - D)** The rate equation is $\text{rate} = k[\text{A}][\text{B}]^2$, the overall order is three, and k has the units $\text{dm}^6/(\text{mol}^2 \cdot \text{s})$.
 - E)** The reaction is first order overall, and k has the units s^{-1} , because each reactant contributes one power.
- 63.** The Haber process, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ with $\Delta H = -92 \text{ kJ/mol}$, uses an iron catalyst. Which statement is correct?
- A)** The catalyst lowers the activation energy of the forward and reverse reactions equally, so equilibrium is reached sooner, but the equilibrium position and ΔH are unchanged.
 - B)** The catalyst raises the activation energy of the reverse reaction, so ammonia accumulates.
 - C)** The catalyst works by raising the average kinetic energy of the gas molecules, so that more collisions are effective.
 - D)** The catalyst makes the reaction more exothermic, which increases the yield at the same temperature.
 - E)** The catalyst shifts the equilibrium position to the right, increasing the final yield of ammonia.

- 64.** Manganese(IV) oxide speeds up the decomposition of hydrogen peroxide, $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$, and the black powder can be recovered unchanged at the end. The best explanation is that MnO_2 :
- A)** is consumed to supply the energy that the reaction needs, which is why a little disappears each time.
 - B)** raises the average kinetic energy of the H_2O_2 molecules, so more collisions exceed the activation energy.
 - C)** shifts the equilibrium of the decomposition towards the products, so the reaction goes faster.
 - D)** reduces the enthalpy change of the decomposition, which releases more energy to drive the reaction.
 - E)** provides an alternative reaction pathway of lower activation energy and is regenerated during the cycle, so it is not consumed.
- 65.** A reaction mixture is heated from 25°C to 55°C and the rate rises sharply. According to collision theory, the main reason is that:
- A)** molecules collide much more often, and this increase in collision frequency alone explains the whole rate rise.
 - B)** faster molecules are more likely to meet in the correct orientation, because high speed corrects the approach angle.
 - C)** the activation energy of the reaction decreases as the temperature rises.
 - D)** a much larger fraction of molecules now has energy at or above the activation energy, so far more collisions are effective.
 - E)** the enthalpy change of the reaction becomes more negative at the higher temperature.

Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

1	C	2	D	3	B	4	E	5	A
6	C	7	A	8	D	9	B	10	E
11	A	12	D	13	B	14	D	15	C
16	A	17	A	18	A	19	C	20	B
21	C	22	E	23	D	24	B	25	D
26	C	27	D	28	C	29	D	30	C
31	E	32	E	33	D	34	E	35	D
36	B	37	E	38	B	39	C	40	A
41	B	42	A	43	E	44	A	45	B
46	E	47	B	48	B	49	D	50	C
51	E	52	A	53	C	54	E	55	A
56	A	57	C	58	C	59	E	60	B
61	B	62	D	63	A	64	E	65	D